



THE UNIVERSITY OF
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FINAL REPORT

The Art of Scientific Computation Power Laws and Finite Size Scaling in the BTW Abelian Sandpiles Model

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Chapter 1

Introduction

1. Introduce the BTW sandpiles model as a model for self-organised criticality and $1/f$ noise [1]
2. Define criticality, self-organised criticality, power law scaling, etc
3. Mathematically describe the BTW model and toppling condition
4. Describe modelling choices for power law and truncated power law distributions

Chapter 2

Simulation

- Describe the simulation procedure including implementation details (numba, burn-in)
- Show plots of basic results found
- Describe the qualitative scaling behaviour observed across different L

Chapter 3

Power Law Fit

1. Linear regression fit and discussion about rigour of this approach
2. Naive Power law (Zeta) fit
3. Zeta with truncated x_{max}

Chapter 4

Finite Size Scaling Effect

1. General form
2. MLE fit for exponentially-truncated power law
3. MLE fit for generalised exp-truncated power law
4. Discussion

Chapter 5

Non-FSS Behaviour of Toppling Number

1. Run-through of previous analysis with the Toppling Number
2. Identify limitations of the analysis

Bibliography

- [1] P. Bak, C. Tang, and K. Wiesenfeld. Self-organized criticality: An explanation of the $1/f$ noise. *Physical review letters*, 59(4):381, 1987.